

Homework 7

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Instructions

Answer the following questions and/or complete the exercises in RMarkdown. Please embed all of your code and push the final work to your repository. Your report should be organized, clean, and run free from errors. Remember, you must remove the `#` for any included code chunks to run.

Load the tidyverse

```
library("tidyverse")  
library("janitor")
```

Data

For this assignment, we will use data from a study on elephants and the effects of poaching on tusk size.

Reference: Chiyo, Patrick I., Vincent Obanda, and David K. Korir. "Illegal tusk harvest and the decline of tusk size in the African elephant." *Ecology and Evolution* 5, 22: 5216–5229 (2015) (<https://doi.org/10.1002/ece3.1769>). Data deposited at Dryad Digital Repository (<https://doi.org/10.5061/dryad.h6t7j>).

1. Before starting data analysis, read the abstract of the paper to get an idea of the questions being asked. In 2-3 sentences, describe what the study is testing and the variables involved.

The study wanted to know if poaching affects elephant tusk size. It looked at tusk size in elephants from 1966-1968 and compared it to those born in the mid-1990s (after severe poaching in the 1970s and 80s). Sex, tusk length, basal tusk circumference, years of collection, shoulder height, and estimated age were all variables of interest.

2. Load `elephants.csv` and store it as a new object called `elephants`.

```
library("tidyverse")  
  
elephants <- read_csv("C:/Users/fluff/Desktop/BIS15L_W26_aehall/lab7/data/elephants.csv")
```

```
## Rows: 777 Columns: 7
## — Column specification —————
## Delimiter: ","
## chr (3): Years of sample collection, Elephant ID, Sex
## dbl (4): Estimated Age (years), shoulder Height in cm, Tusk Length in cm, T...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

3. Clean the data by converting variable names to lowercase with no spaces or special characters.

```
library("janitor")

elephants <- elephants %>% #clean names
  clean_names()
names(elephants) #check names
```

```
## [1] "years_of_sample_collection" "elephant_id"
## [3] "sex"                        "estimated_age_years"
## [5] "shoulder_height_in_cm"     "tusk_length_in_cm"
## [7] "tusk_circumference_in_cm"
```

4. Use one or more of the summary functions you have learned to get an idea of the structure of the data.

```
glimpse(elephants)
```

```
## Rows: 777
## Columns: 7
## $ years_of_sample_collection <chr> "1966-68", "1966-68", "1966-68", "1966-68",...
## $ elephant_id               <chr> "12", "34", "162", "292", "11", "152", "264...
## $ sex                       <chr> "f", "f", "f", "f", "f", "f", "f", "f", "f"...
## $ estimated_age_years       <dbl> 0.080, 0.080, 0.083, 0.083, 0.250, 0.250, 0...
## $ shoulder_height_in_cm     <dbl> 102, 89, 89, 92, 133, 100, 93, 108, 108, 12...
## $ tusk_length_in_cm         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
## $ tusk_circumference_in_cm  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
```

5. Use `mutate()` Change the variables `years_of_sample_collection`, `elephant_id`, and `sex` to factors. Be sure to store the output as a new dataframe and use it for the remaining questions.

```

elephants <- elephants %>%
  mutate(across(c(years_of_sample_collection, elephant_id, sex), as.factor))

glimpse(elephants) #check work

```

```

## Rows: 777
## Columns: 7
## $ years_of_sample_collection <fct> 1966-68, 1966-68, 1966-68, 1966-68, 1966-68...
## $ elephant_id               <fct> 12, 34, 162, 292, 11, 152, 264, 263, 266, 2...
## $ sex                       <fct> f, f, f, f, f, f, f, f, f, f, f, f, f, f, f...
## $ estimated_age_years       <dbl> 0.080, 0.080, 0.083, 0.083, 0.250, 0.250, 0...
## $ shoulder_height_in_cm     <dbl> 102, 89, 89, 92, 133, 100, 93, 108, 108, 12...
## $ tusk_length_in_cm         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
## $ tusk_circumference_in_cm  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...

```

6. From which years were data collected? Show the sample periods below.

```
levels(elephants$years_of_sample_collection)
```

```
## [1] "1966-68" "2005-13"
```

7. How many males and females were sampled in this study?

```

nfemales <- elephants %>% #count number of females
  filter(sex=="f") %>%
  nrow()

nmales <- elephants %>% #count number of males
  filter(sex=="m") %>%
  nrow()

#There are 416 females & 361 males in the study.

```

8. What is the mean, median, and standard deviation for age of males and females included in the study? Separate the results by year of sample collection. Does the sampling look even between years and sexes?

```

elephants %>%
  filter(sex=="f" & years_of_sample_collection=="1966-68") %>%
  select(estimated_age_years, sex) %>%
  summarize(mean_age_f_1966t68=mean(estimated_age_years, na.rm=T),
            median_age_f_1966t68=median(estimated_age_years, na.rm=T),
            sd_age_f_1966t68=sd(estimated_age_years, na.rm=T),
            total=n())

```

```

## # A tibble: 1 × 4
##   mean_age_f_1966t68 median_age_f_1966t68 sd_age_f_1966t68 total
##           <dbl>           <dbl>           <dbl> <int>
## 1             17.6             15             13.6    323

```

```

elephants %>%
  filter(sex=="f" & years_of_sample_collection=="2005-13") %>%
  select(estimated_age_years, sex) %>%
  summarize(mean_age_f_2005t13=mean(estimated_age_years, na.rm=T),
            median_age_f_2005t13=median(estimated_age_years, na.rm=T),
            sd_age_f_2005t13=sd(estimated_age_years, na.rm=T),
            total=n())

```

```

## # A tibble: 1 × 4
##   mean_age_f_2005t13 median_age_f_2005t13 sd_age_f_2005t13 total
##           <dbl>           <dbl>           <dbl> <int>
## 1             17.9             17.5             11.0    93

```

```

elephants %>%
  filter(sex=="m" & years_of_sample_collection=="1966-68") %>%
  select(estimated_age_years, sex) %>%
  summarize(mean_age_m_1966t68=mean(estimated_age_years, na.rm=T),
            median_age_m_1966t68=median(estimated_age_years, na.rm=T),
            sd_age_m_1966t68=sd(estimated_age_years, na.rm=T),
            total=n())

```

```

## # A tibble: 1 × 4
##   mean_age_m_1966t68 median_age_m_1966t68 sd_age_m_1966t68 total
##           <dbl>           <dbl>           <dbl> <int>
## 1             10.8             8             9.19    282

```

```

elephants %>%
  filter(sex=="m" & years_of_sample_collection=="2005-13") %>%
  select(estimated_age_years, sex) %>%
  summarize(mean_age_m_2005t13=mean(estimated_age_years, na.rm=T),
            median_age_m_2005t13=median(estimated_age_years, na.rm=T),
            sd_age_m_2005t13=sd(estimated_age_years, na.rm=T),
            total=n())

```

```

## # A tibble: 1 × 4
##   mean_age_m_2005t13 median_age_m_2005t13 sd_age_m_2005t13 total
##           <dbl>           <dbl>           <dbl> <int>
## 1             16.7             9             13.9     79

```

```

#The sampling looks pretty even between sexes.
#There are significantly fewer elephants in the 2005-13 sampling.

```

9. Is age (independent variable) a positive predictor of tusk length (dependent variable)? Create a plot that shows the relationship between these variables and add a linear model fit line.

```

ggplot(data=elephants,
       mapping=aes(x=estimated_age_years, y=tusk_length_in_cm, color=years_of_sample_
collection))+
  geom_point(na.rm=T)+
  geom_smooth(method=lm, se=T)+
  labs(x="Estimated Age (yrs)", y="Tusk Length (cm)", title="Estimated Age vs Tusk Le
ngth by Collection Period")

```

```

## `geom_smooth()` using formula = 'y ~ x'

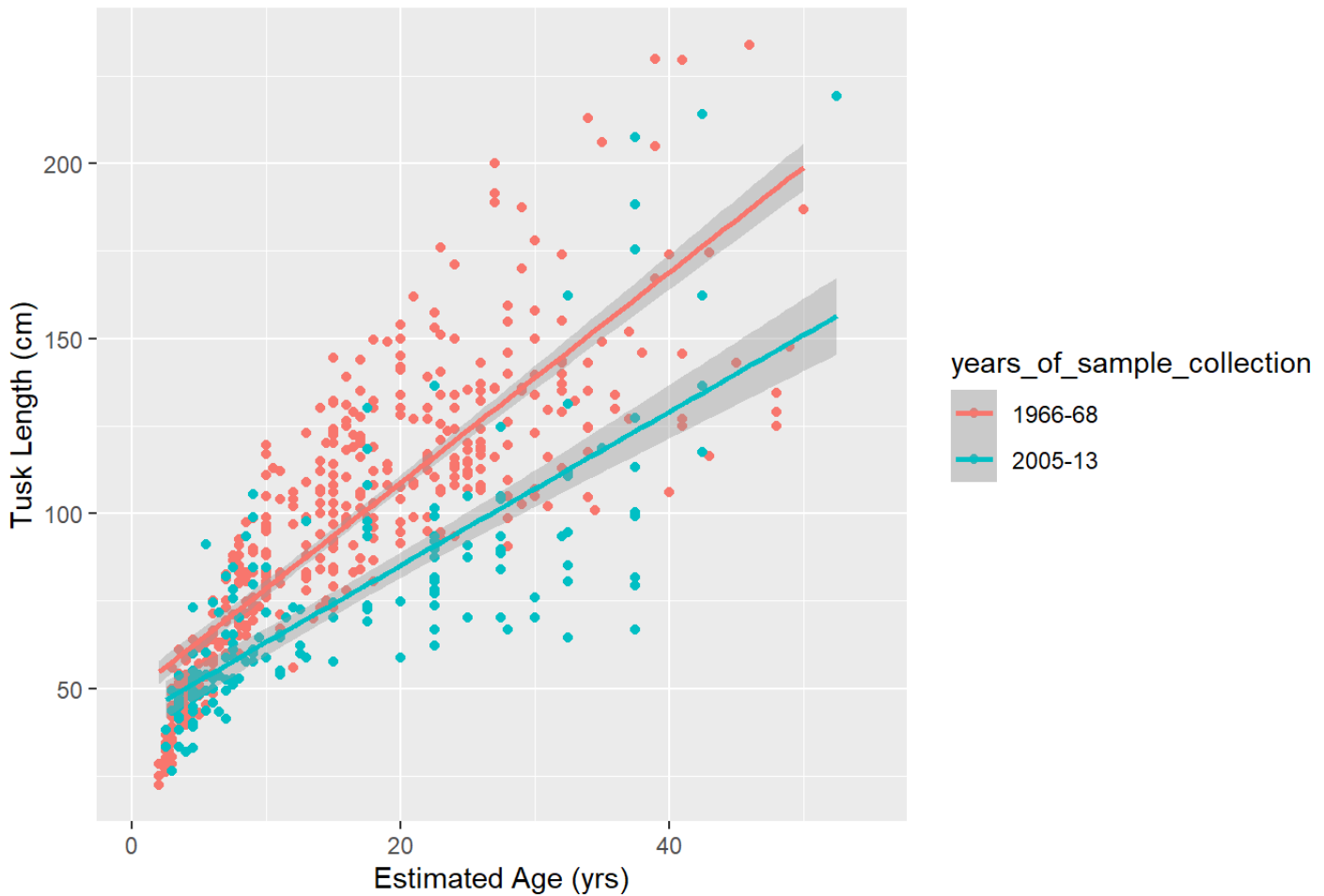
```

```

## Warning: Removed 182 rows containing non-finite outside the scale range
## (`stat_smooth()`).

```

Estimated Age vs Tusk Length by Collection Period



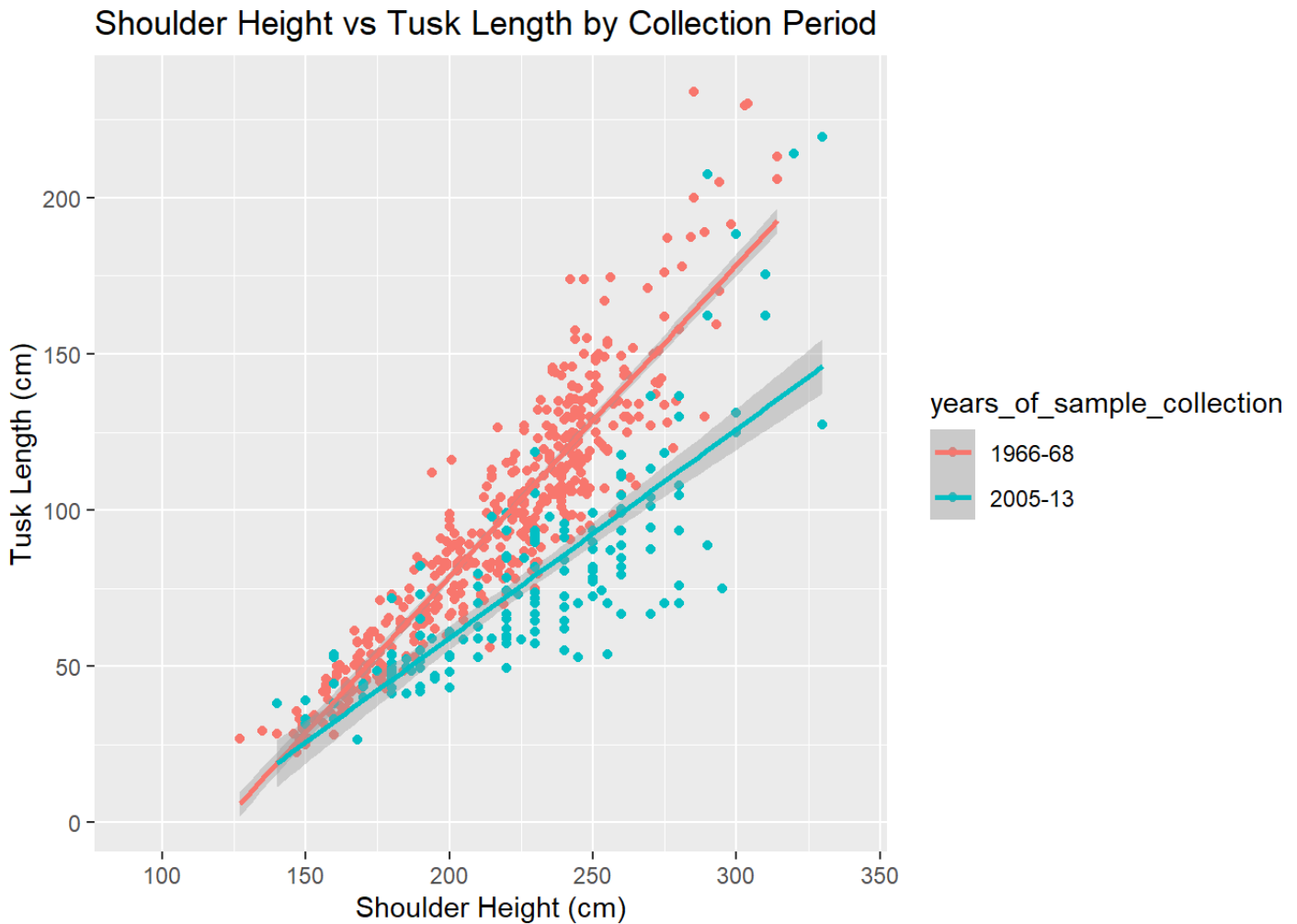
#Somewhat, yes.

10. Is shoulder height (independent variable) a positive predictor of tusk length (dependent variable)? Create a plot that shows the relationship between these variables and add a linear model fit line.

```
ggplot(data=elephants,
       mapping=aes(x=shoulder_height_in_cm, y=tusk_length_in_cm, color=years_of_sample_collection))+
  geom_point(na.rm=T)+
  geom_smooth(method=lm, se=T)+
  labs(x="Shoulder Height (cm)", y="Tusk Length (cm)", title="Shoulder Height vs Tusk Length by Collection Period")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 181 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```



```
#Yes, mostly.
```

11. The authors argue that because poachers preferentially target elephants with large tusks, this has resulted in a decrease in average tusk length. Is this supported by the data? Show your code and calculations below.

```
elephants1966 <- elephants %>%
  filter(years_of_sample_collection=="1966-68")
elephants2005 <- elephants %>%
  filter(years_of_sample_collection=="2005-13")

elephants1966 %>%
  summarize(mean_tusk_length_1966t68=mean(tusk_length_in_cm, na.rm=T),
            total=n())
```

```
## # A tibble: 1 × 2
##   mean_tusk_length_1966t68 total
##               <dbl> <int>
## 1              96.9    605
```

```
elephants2005 %>%
  summarize(mean_tusk_length_2005t13=mean(tusk_length_in_cm, na.rm=T),
            total=n())
```

```
## # A tibble: 1 × 2
##   mean_tusk_length_2005t13 total
##               <dbl> <int>
## 1              77.2    172
```

```
#Yes, this is supported by the data.
```

12. Male elephants reach effective sexual maturity at 25 years while females are sexually mature at 12 years. Make a new dataframe that extracts only the males and females at sexual maturity. Then, make a plot that shows the range of tusk length between the two sample periods for these mature elephants.

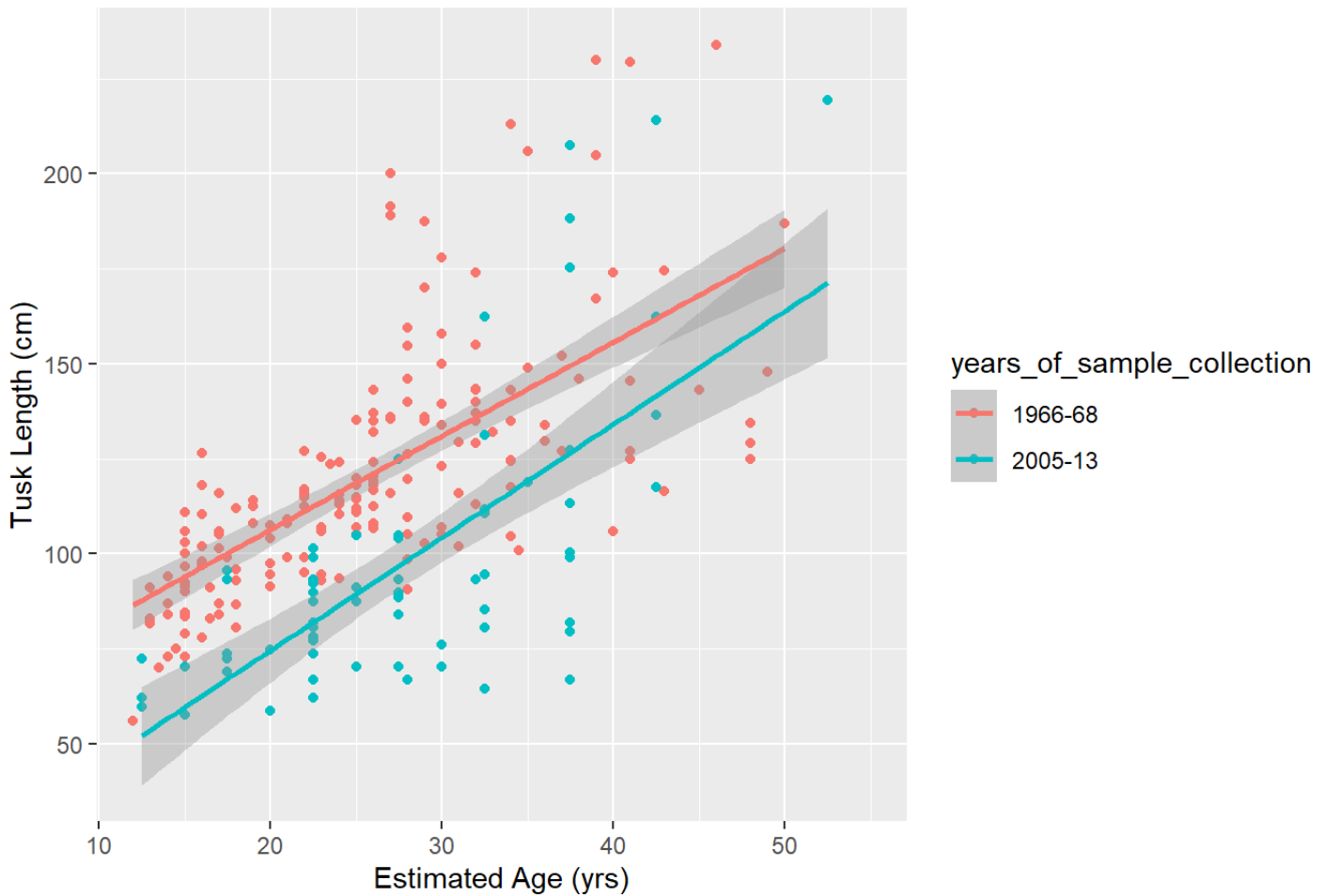
```
mature_elephants <- elephants %>%
  filter((sex=="f" & estimated_age_years>=12) | (sex=="m" & estimated_age_years>=25))

ggplot(data=mature_elephants,
       mapping=aes(x=estimated_age_years, y=tusk_length_in_cm, color=years_of_sample_
collection))+
  geom_point(na.rm=T)+
  geom_smooth(method=lm, se=T)+
  labs(x="Estimated Age (yrs)", y="Tusk Length (cm)", title="Mature Elephants: Estima
ted Age vs Tusk Length by Collection Period")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 44 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```


Mature Elephants: Estimated Age vs Tusk Length by Collection Period



Submit the Homework

1. Save your work and knit the .rmd file.
2. Open the .html file and "print" it to a .pdf file in Google Chrome (not Safari).
3. Go to the class Canvas page and open Gradescope.
4. Submit your .pdf file to the homework assignment- be sure to assign the pages to the correct questions.
5. Commit and push your work to your repository.